

Each subject has 6 exercises: 4 operations, a conversion and exercise 6 related to the internal representations.

Subject

Perform the following operations:

(0.75p) 1. $33221100_{(4)} + 123032122_{(4)} = ?_{(4)}$

(0.75p) 2. $1230056_{(7)} - 445566_{(7)} = ?_{(7)}$

(0.75p) 3. $ABCDE1_{(16)} * 7_{(16)} = ?_{(16)}$

(0.75p) 4. $54321_{(6)} : 3_{(6)} = ?_{(6)} \text{ remainder } ?_{(6)}$

!! 0.25p from 0.75p are for writing the calculations for at least 5 iterations.

(3p) 2.25p for the calculations needed in the conversion and 0.75p for theory (explain why do you choose a specific method, its advantage, describe the method).

Examples of possible exercises:

5. Convert the number 12345,04 from base 6 into base 8 with 2 digits at the fractional part. Explain the chosen method.
5. Convert the number ABCDE,1234 from base 16 into base 4 and the number 11223300 from base 4 into base 8. Explain the chosen method.
5. Convert the number 45078,63 from base 9 into base 5 with 2 digits at the fractional part. Explain the chosen method.

(3p) 2.25p for the calculations needed in the representation and 0.75p for theory (explain the representations, formal definitions, advantages-disadvantages, practical rules).

Examples of possible exercises:

6. Floating point representation of real numbers: theory.
Represent in single precision with a mantissa >1 the number -3456,78.
6. Codes (direct, inverse, complementary) for signed subunitary numbers: theory and example for the number -0,241, representation on 16 bits.
6. Addition and subtraction in complementary code for signed integers: theory.
Examples for $55+91$, $55-91$ and $-91-55$, representation on 8 bits.
6. Fixed point representation of real numbers: theory.
Represent in fixed point notation on 32 bits the number -4501,33.
6. Floating point representation of real numbers: theory.
Represent in single precision with a subunitary mantissa the number 0,023.
6. Codes (direct, inverse, complementary) for signed integers: theory and example for the number 12345, representation on 16 bits.
6. Addition and subtraction in complementary code for subunitary numbers: theory.
Examples for $-\frac{7}{16} - 0,71$ and $\frac{7}{16} - 0,71$, representation on 16 bits.
6. Find the real number x having B0CDA000₍₁₆₎ as its fixed-point representation on 32 bits with I=18 bits and F=13 bits.
6. Find the real number x having B0CDA000₍₁₆₎ as its floating-point representation, SP with $m>1$.